

Sources of the Magnetic Field

Currents make magnetic fields: the Biot–Savart law, Ampère's law, solenoids, magnetic materials, and the giant magnets of MRI.

By **Pratik Bhanuse**, PhD Physics candidate, Western University · pratikbhanuse.github.io

In 1820 Hans Christian Ørsted noticed that a current in a wire deflected a nearby compass needle. Within weeks, Biot, Savart and Ampère had worked out the laws: **moving charges create magnetic fields**, just as charges create electric fields. This chapter finds the fields of wires, loops and coils, using the Biot–Savart law (the magnetic counterpart of Coulomb's law) and Ampère's law (the counterpart of Gauss's law). It explains how solenoids produce the uniform fields used in MRI, why parallel currents attract (which once defined the ampere), and how magnetic materials such as iron multiply fields a thousandfold.

YOU WILL BE ABLE TO

- Use the Biot–Savart law to find the fields of straight wires, arcs and circular loops.
- Calculate the force between parallel currents, and explain the definition of the ampere.
- State Ampère's law and use it for long wires, coaxial cables, solenoids and toroids.
- Describe the field of a magnetic dipole and of Helmholtz coils.
- Explain diamagnetism, paramagnetism and ferromagnetism, hysteresis and magnetic circuits.
- Apply these ideas to MRI magnets, transcranial magnetic stimulation and magnetic shielding.

THINK FIRST

An MRI scanner's 3 T field is produced by a coil of wire carrying a few hundred amperes. Once it is “ramped up”, the power supply is disconnected, yet the field stays on for years. How can a current flow for years with no battery?

The coil is made of a **superconductor**, cooled by liquid helium to about 4 K. Below its critical temperature it has exactly zero resistance, so once a current is established it flows round the closed loop with no energy loss. The magnet is “persistent”. The danger is a **quench**: if part of the coil warms above its critical temperature, it suddenly gains resistance, and the magnetic energy stored in the field, several megajoules, is converted to heat in seconds. That boils off the helium violently, which is why scanners have quench pipes to vent it outdoors.

§19.1

The Biot–Savart law

A short segment of wire $d\vec{l}$ carrying current I produces, at a point displaced by $\vec{r}$ from the segment, a magnetic field

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}, \quad \mu_0 = 4\pi \times 10^{-7} \text{ T m/A.}$$

The constant μ_0 is the **permeability of free space**. Since the 2019 SI redefinition its value is measured rather than exact, but it agrees with $4\pi \times 10^{-7}$ to about one part in 10^{10} .

Features of the Biot–Savart law:

- The field falls off as $1/r^2$, like Coulomb's law.
- $d\vec{B}$ is **perpendicular** to both $d\vec{l}$ and $\hat{r}$, so the field circles around the current.
- A segment produces no field along its own line, where $d\vec{l} \times \hat{r} = 0$.

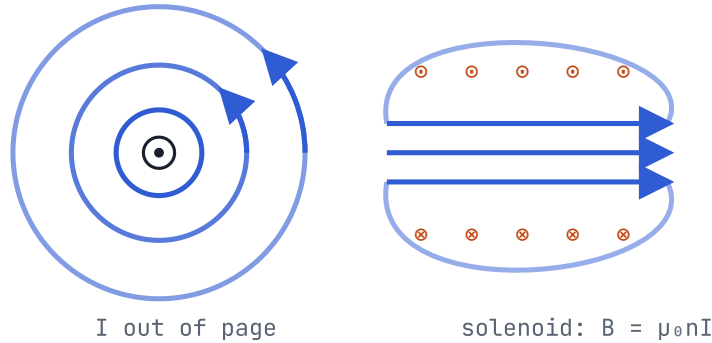
The total field is the vector sum of the contributions from every segment: $\vec{B} = \int d\vec{B}$.

The long straight wire

For an infinitely long straight wire, the Biot–Savart integral (Problem P19.13, with both ends at infinity) gives

$$B = \frac{\mu_0 I}{2\pi r}.$$

The field lines are **circles** centred on the wire. Their direction follows the **right-hand grip rule**: grip the wire with your right thumb along the current, and your fingers curl in the direction of $\vec{B}$.



Left: the circular field lines of a long straight wire carrying current out of the page. The field weakens as $1/r$. Right: a solenoid produces a strong, nearly uniform field inside, and a weak field outside.

WORKED EXAMPLE 19.1

A wire next to a compass

A long straight wire carries 10 A. Find the magnetic field 5.0 cm away, and compare it with the Earth's field of about $50 \mu\text{T}$.

Solution

$$B = \frac{\mu_0 I}{2\pi r} = \frac{(4\pi \times 10^{-7})(10)}{2\pi(0.050)} = 4.0 \times 10^{-5} \text{ T} = 40 \mu\text{T}.$$

This is comparable with the Earth's field, so a compass placed near such a wire would be strongly deflected. That is exactly what Ørsted saw. It is also why aircraft and ship compasses are kept away from current-carrying cables.

The circular loop

Every element of a loop of radius R is the same distance from the centre, and its contribution there points along the axis. So at the **centre**:

$$B = \frac{\mu_0 I}{2R}, \quad \text{or} \quad B = \frac{\mu_0 N I}{2R} \text{ for a flat coil of } N \text{ turns.}$$

On the **axis**, at a distance x from the centre, the components perpendicular to the axis cancel by symmetry (just as for the ring of charge in Chapter 13), and

$$B = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}}.$$

Far away ($x \gg R$), $B \approx \frac{\mu_0}{2\pi} \frac{\mu}{x^3}$, where $\mu = I\pi R^2$ is the magnetic moment. This is a **magnetic dipole** field, falling off as $1/r^3$, with the same pattern as the electric dipole field of Chapter 13. Bar magnets, the Earth, and individual atoms all produce dipole fields at large distances.

A **circular arc** that subtends an angle ϕ (in radians) at its centre contributes a field $B = \frac{\mu_0 I \phi}{4\pi R}$ there. A full circle has $\phi = 2\pi$, which gives $\mu_0 I / 2R$ again.

WORKED EXAMPLE 19.2

A flat coil

A flat coil of 50 turns and radius 5.0 cm carries 2.0 A. Find the field at its centre, and on its axis 5.0 cm from the centre.

Solution

At the centre:

$$B_0 = \frac{\mu_0 N I}{2R} = \frac{(4\pi \times 10^{-7})(50)(2.0)}{2(0.050)} = 1.3 \times 10^{-3} \text{ T}.$$

On the axis at $x = R$:

$$B = B_0 \frac{R^3}{(2R^2)^{3/2}} = \frac{B_0}{2\sqrt{2}} = 4.4 \times 10^{-4} \text{ T}.$$

§19.2

Force between parallel currents

A long wire carrying current I_1 produces a field $B_1 = \mu_0 I_1 / (2\pi d)$ at a parallel wire a distance d away. The second wire, carrying I_2 , then feels a force per unit length

$$\frac{F}{L} = I_2 B_1 = \frac{\mu_0 I_1 I_2}{2\pi d}.$$

KEY IDEA

Parallel currents attract. Antiparallel currents repel. This is the opposite of electric charges, where like charges repel.

Until 2019, the ampere was *defined* by this force: two infinitely long, thin wires 1 m apart, each carrying 1 A, attract with exactly 2×10^{-7} N per metre. (The ampere is now defined by fixing the elementary charge e .)

WORKED EXAMPLE 19.3**Forces in a busbar during a fault**

Two parallel busbars in a substation are 0.50 m apart. During a short-circuit fault, each carries 20 kA, in opposite directions. Find the force per metre on each bar.

Solution

$$\frac{F}{L} = \frac{(4\pi \times 10^{-7})(2.0 \times 10^4)^2}{2\pi(0.50)} = \frac{(2.0 \times 10^{-7})(4.0 \times 10^8)}{0.50} = 160 \text{ N/m},$$

and the bars **repel**. Over a 5 m span, that is 800 N, applied suddenly. Busbar supports must be designed to withstand these short-circuit forces, which grow as the *square* of the current.

§19.3

Ampère's law

For any closed path (an “Amperian loop”), the line integral of $\vec{B}$ around the path equals μ_0 times the current passing through it:

KEY IDEA

Ampère's law:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}.$$

Currents passing through the loop count positive if they follow the right-hand rule relative to the direction in which the loop is traversed, and negative otherwise. Currents outside the loop contribute nothing to the integral.

Ampère's law plays the role for magnetism that Gauss's law plays for electricity. It is always true for steady currents. It is *useful* only when symmetry makes $\vec{B}$ constant along the path and either parallel or perpendicular to it. Chapter 21 shows how Maxwell extended it to changing fields.

Long straight wire, inside and outside

Use a circular Amperian loop of radius r , centred on the wire. By symmetry, $\vec{B}$ is tangent to the loop and the same everywhere on it, so $\oint \vec{B} \cdot d\vec{l} = B(2\pi r)$.

- Outside ($r > R$): $I_{\text{enc}} = I$, so $B = \frac{\mu_0 I}{2\pi r}$. ✓ (This agrees with Biot-Savart.)
- Inside a wire of radius R with uniform current density: $I_{\text{enc}} = I \frac{r^2}{R^2}$, so $B = \frac{\mu_0 I r}{2\pi R^2}$.

The field rises linearly inside the wire, peaks at its surface, then falls as $1/r$, the same shape as the electric field of a uniformly charged cylinder.

Coaxial cable. The inner and outer conductors carry equal and opposite currents. A loop outside the cable encloses zero net current, so $\vec{B} = 0$ outside. The cable neither radiates magnetic fields nor picks them up.

The solenoid

A **solenoid** is a long, tightly wound helical coil with n turns per unit length. Inside, far from the ends, the field is uniform and parallel to the axis. Outside, it is nearly zero. Take a rectangular Amperian loop with one side of length ℓ inside the solenoid, parallel to the axis, and the opposite side outside:

- Only the inside side contributes to $\oint \vec{B} \cdot d\vec{l}$, giving $B\ell$. The outside side has $B \approx 0$, and the two ends are perpendicular to $\vec{B}$.
- The loop encloses $n\ell$ turns, so $I_{\text{enc}} = n\ell I$.

$$B\ell = \mu_0 n \ell I \Rightarrow B = \mu_0 n I.$$

The field does not depend on the solenoid's radius, or on where you are inside it. At each end, the field falls to half of this value.

The toroid

A **toroid** is a solenoid bent round into a doughnut, with N turns in total. Using a circular Amperian loop of radius r inside the windings:

$$B = \frac{\mu_0 N I}{2\pi r}.$$

Outside the toroid, $B = 0$. The field is completely confined, which makes toroids ideal for transformers and inductors that must not interfere with nearby circuits.

WORKED EXAMPLE 19.4

An MRI magnet

A whole-body MRI magnet is a superconducting solenoid 1.5 m long and 0.90 m in diameter, carrying 400 A, that produces 3.0 T. (a) How many turns per metre, and in total, does it need? (b) The energy density of a magnetic field is $B^2/2\mu_0$ (Chapter 20). Estimate the energy stored inside the bore.

Solution

(a) Turns:

$$n = \frac{B}{\mu_0 I} = \frac{3.0}{(4\pi \times 10^{-7})(400)} = 6.0 \times 10^3 \text{ turns/m}, \quad N = nL \approx 9 \times 10^3 \text{ turns}.$$

(b) Energy:

$$u = \frac{B^2}{2\mu_0} = \frac{9.0}{2(4\pi \times 10^{-7})} = 3.6 \times 10^6 \text{ J/m}^3.$$

The bore's volume is $\pi(0.45)^2(1.5) = 0.95 \text{ m}^3$, so

$$U \approx 3.4 \text{ MJ}.$$

That is about the energy of a 1-tonne car at 300 km/h. In a quench, it is all released within seconds as heat.

WORKED EXAMPLE 19.5

A toroidal coil

A toroid with 500 turns and a mean radius of 10 cm carries 2.0 A. Find the field inside it.

Solution

$$B = \frac{\mu_0 NI}{2\pi r} = \frac{(4\pi \times 10^{-7})(500)(2.0)}{2\pi(0.10)} = 2.0 \times 10^{-3} \text{ T.}$$

§19.4

Magnetic materials

Every atom contains moving electrons, with orbital motion and intrinsic **spin**, and so acts as a tiny magnetic dipole. The natural unit of atomic magnetic moment is the **Bohr magneton**, $\mu_B = 9.27 \times 10^{-24} \text{ A m}^2$ (Problem P19.18). How these atomic moments respond to an applied field divides materials into three classes. The response is measured by the **relative permeability** μ_r : inside a material that fills the region, the field becomes $B = \mu_r \mu_0 nI$.

TYPE	μ_r	MECHANISM	EXAMPLES
Diamagnetic	slightly < 1 (by about 10^{-5})	Induced orbital currents oppose the applied field. Present in all matter.	Water, tissue, copper, bismuth. Superconductors are perfect diamagnets ($\mu_r = 0$).
Paramagnetic	slightly > 1 (by 10^{-5} – 10^{-3})	Permanent atomic moments partly align with the field, against thermal disorder.	Aluminium, oxygen, gadolinium MRI contrast agents, deoxyhaemoglobin
Ferromagnetic	10^2 – 10^5	Strong quantum “exchange” coupling lines up neighbouring moments within	Iron, nickel, cobalt, many alloys

TYPE		MECHANISM	EXAMPLES
		domains. An applied field grows the favourably aligned domains.	

Ferromagnets show **hysteresis**: their magnetisation depends on their history. They can retain magnetisation after the field is removed, which makes permanent magnets, and they *saturate* at high fields, typically around 1.5–2 T for iron. Above the **Curie temperature** (1043 K for iron), thermal agitation destroys the domain alignment, and the material becomes paramagnetic.

- *Soft* magnetic materials, such as silicon steel, have narrow hysteresis loops and are easy to magnetise and demagnetise, so they are used in transformer cores.
- *Hard* materials, such as NdFeB, have wide loops and are used for permanent magnets, in motors, headphones and wind turbines.

IN PRACTICE

Magnetic materials in medicine and engineering.

- **MRI contrast agents** contain paramagnetic gadolinium. Their large atomic moments speed up the relaxation of nearby protons, brightening blood vessels and tumours on the image.
- **Functional MRI (fMRI)** exploits the fact that *deoxygenated* haemoglobin is paramagnetic while oxygenated haemoglobin is diamagnetic. Active brain regions receive more oxygenated blood, which subtly changes the local field and the signal: the BOLD effect.
- **Magnetic shielding.** Rooms for magnetoencephalography (MEG) are lined with high-permeability “mu-metal”, which channels the Earth’s field and urban magnetic noise *around* the room, reducing it by a factor of 10^3 – 10^5 .
- **Transformers and motors** use iron cores to multiply fields by $\mu_r \sim 10^3$, so that strong fields can be produced with modest currents (Problem P19.17).

WORKED EXAMPLE 19.6

Transcranial magnetic stimulation

A TMS coil used to stimulate the brain has 15 turns of radius 3.5 cm and carries a peak current of 5.0 kA in a pulse lasting about 0.1 ms. Find the peak field at the centre of the coil.

Solution

$$B = \frac{\mu_0 NI}{2R} = \frac{(4\pi \times 10^{-7})(15)(5.0 \times 10^3)}{2(0.035)} = 1.3 \text{ T}.$$

Evaluate. The field itself does not stimulate neurons. What matters is its rapid *change*, from 0 to 1.3 T in about 0.1 ms. By Faraday's law (Chapter 20), the changing field induces electric currents in the brain tissue beneath the coil, and these can trigger neurons to fire. TMS is approved for treating depression that does not respond to drugs, and it is used to map the motor cortex before brain surgery.

CHAPTER SUMMARY

- Biot-Savart: $d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$, with $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$.
- Long wire: $B = \mu_0 I / 2\pi r$, with circular field lines (right-hand grip rule). Loop centre: $\mu_0 NI / 2R$. Loop axis: $\mu_0 I R^2 / [2(x^2 + R^2)^{3/2}]$. Arc: $\mu_0 I \phi / 4\pi R$.
- Parallel wires: $F/L = \mu_0 I_1 I_2 / 2\pi d$. Parallel currents attract, antiparallel currents repel.
- Ampère: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$. Inside a wire: $\mu_0 I r / 2\pi R^2$. Solenoid: $\mu_0 n I$. Toroid: $\mu_0 NI / 2\pi r$. Coaxial cable: 0 outside.
- A distant loop gives a dipole field $\propto \mu / r^3$, with $\mu = IA$.
- Materials: diamagnetic ($\mu_r < 1$), paramagnetic ($\mu_r > 1$, small effect), ferromagnetic ($\mu_r \gg 1$, with domains, hysteresis, saturation and a Curie temperature).

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P19.1

Explain why two parallel wires carrying currents in the same direction attract each other, using the field of one wire and the force on the other.

P19.2

Why is the field inside a long solenoid nearly uniform, and the field outside nearly zero?

P19.3

Ampère's law is true for any closed loop. Why can't it be used to find the field of a short, straight segment of wire?

P19.4

A permanent magnet is heated above its Curie temperature, then cooled again with no external field present. Is it still a magnet? Explain using the idea of domains.

LEVEL B – STANDARD PROBLEMS**P19.5**

Two long parallel wires 10 cm apart carry 5.0 A and 8.0 A in the same direction. Find the magnetic field midway between them, and the force per metre between the wires.

P19.6

For the wires in P19.5, find the point between them where the magnetic field is zero.

P19.7

An overhead power line 15 m above the ground carries 500 A. Find the magnetic field at ground level directly below it, and compare it with the Earth's field. (International guidelines limit public exposure to about 200 μT at 50 Hz.)

P19.8

A circular coil of 100 turns and radius 10 cm carries 1.0 A. Find the field at its centre, and on its axis 10 cm from the centre.

P19.9

A solenoid 50 cm long has 1000 turns and carries 2.0 A. Find the field inside it, and the magnetic flux through a cross-section of area 4.0 cm^2 .

P19.10

A coaxial cable carries 5.0 A in its inner conductor (radius 1.0 mm) and 5.0 A back in its outer conductor (inner radius 4.0 mm). Find the magnetic field at 0.50 mm, 2.0 mm and 10 mm from the axis.

P19.11

An electron travels at 1.0×10^6 m/s parallel to a long straight wire carrying 20 A, at a distance of 2.0 cm, moving in the same direction as the current. Find the magnitude and direction of the force on it.

P19.12

TMS coil. For the coil of Example 19.6, the current rises from zero to 5.0 kA in 0.10 ms. (a) Find the average rate of change of the field at the centre. (b) What is the force per metre between two adjacent turns 4.0 mm apart, at peak current? Why must TMS coils be built so robustly?

LEVEL C – CHALLENGE PROBLEMS**P19.13**

Use the Biot–Savart law to show that a straight wire segment carrying current I produces a field $B = \frac{\mu_0 I}{4\pi d} (\sin \alpha_2 - \sin \alpha_1)$ at a perpendicular distance d . Here α_1 and α_2 are the angles, measured from the perpendicular, to the two ends of the segment. Use your result to (a) recover the infinite-wire result and (b) find the field at the centre of a square loop of side a carrying current I . Compare (b) with the field at the centre of a circular loop of the same perimeter.

P19.14

A long wire is bent into a hairpin: two long parallel straight sections joined by a semicircle of radius R . Find the magnetic field at the centre of the semicircle.

P19.15

Helmholtz coils. Two identical coaxial coils of radius R and N turns, carrying the same current I in the same direction, are separated by a distance equal to R . (a) Show that the field at the midpoint is $B = (4/5)^{3/2} \mu_0 NI / R$. (b) Show that dB/dx and d^2B/dx^2 are both zero at the midpoint, so the field there is extremely uniform. (c) Evaluate B for $R = 0.30$ m, $N = 100$ and $I = 2.0$ A. Give two uses of Helmholtz coils.

P19.16

A long cylindrical conductor of radius R carries a current whose density increases with radius: $J = J_0 r / R$. (a) Find the total current. (b) Use Ampère's law to find $B(r)$ inside and outside the conductor.

P19.17

Magnetic circuit with an air gap. An iron toroid with a mean path length of 0.30 m and $\mu_r = 2000$ has a 2.0 mm air gap cut through it. It is wound with 500 turns carrying 2.0 A. (a) Using Ampère's law around the core and the gap, together with the fact that B is nearly the same in the iron and the gap, show that $B = \mu_0 NI / (g + L/\mu_r)$. (b) Evaluate B in the gap. (c) Compare it with the field the same winding would produce with no gap. Comment on the result, given that iron saturates at about 2 T.

P19.18

The Bohr magneton. In the Bohr model of hydrogen, the electron moves in a circle of radius 5.29×10^{-11} m at 2.19×10^6 m/s. (a) Find the equivalent current. (b) Find the magnetic moment of the orbit, and show that it equals $\mu_B = e\hbar/2m_e = 9.27 \times 10^{-24}$ A m². (c) Find the magnetic field the orbiting electron produces at the proton. Comment on its size.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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